

A
Capstone Project Synopsis
On
Resum.AI

Submitted during 6th semester in partial fulfilment of the requirements for the award of degree of

**Bachelor of Technology
Computer Engineering
(with specialisation in Data Science)**

by

Abhishek Verma
(22001016002)

Under supervision of
Dr. Sonali Gupta



Department of Computer Engineering

**J.C. BOSE UNIVERSITY OF SCIENCE &
TECHNOLOGY, YMCA
FARIDABAD-121006
May-2025**

1. Introduction:

In the competitive landscape of modern recruitment, efficiently analyzing and shortlisting resumes is a significant challenge for HR departments and recruiters. Traditional resume screening methods are time-consuming, prone to human bias, and often inefficient when handling large volumes of applications. To address these issues, we propose the development of **resum.AI**, a smart resume analyzer that leverages modern AI technologies to automate and enhance the resume evaluation process.

resum.AI is designed to provide a seamless, intelligent, and scalable solution for analyzing resumes in real-time. The system utilizes **Google Gemini API** for natural language understanding and **Streamlit** as a lightweight web interface, backed by a **SQLite3 database** for data management. Users can upload resumes and job descriptions, and the system provides match scores, skill-gap analysis, and personalized improvement suggestions—making it a powerful tool for both recruiters and job seekers.

2. Objective of the Project:

The main objective of **resum.AI** is to automate the process of analyzing and evaluating resumes using artificial intelligence, thereby reducing manual effort and enhancing decision-making accuracy. This system aims to:

- Streamline the resume screening process using AI.
 - Provide a **match percentage** between resume content and job descriptions.
 - Highlight **key strengths** and **missing skills** in a resume.
 - Offer **actionable feedback** for resume improvement.
 - Ensure secure and organized data storage using **SQLite3**.
 - Present results through a **clean, interactive UI** using Streamlit.
 - Support scalability for use in educational institutions, HR firms, and online job platforms.
-

3. Tools Required / Platform Used:

- **Tools:** VS Code, Google Gemini API, Streamlit, SQLite3
- **Languages/Technologies:** Python, HTML5, CSS, JavaScript
- **Libraries/Frameworks:** fitz, docx, sqlite3, Streamlit

4. Conclusion:

In conclusion, **resum.AI** represents a significant step forward in the field of automated resume evaluation. By leveraging powerful AI capabilities through Google Gemini and integrating with an intuitive frontend via Streamlit, the system offers a smart, efficient, and scalable solution for modern recruitment needs. With features like job match scoring, improvement recommendations, and resume parsing, it not only streamlines administrative processes but also empowers users to optimize their career documents effectively. Furthermore, its lightweight architecture, built with SQLite3 and Python, ensures smooth deployment and adaptability for future enhancements. Overall, **resum.AI** delivers a valuable contribution to both the recruitment ecosystem and the personal development of job seekers.

Brief Profile Of the Team :

Name: AKUL YADAV

Roll No: 22001016007

Email: 22001016007@jcboseust.ac.in

Mobile: 9817846707

Name: ABHISHEK VERMA

Roll No.:22001016002

Email : 22001016002@jcboseust.ac.in

Mobile : 9306928421